

Reflections

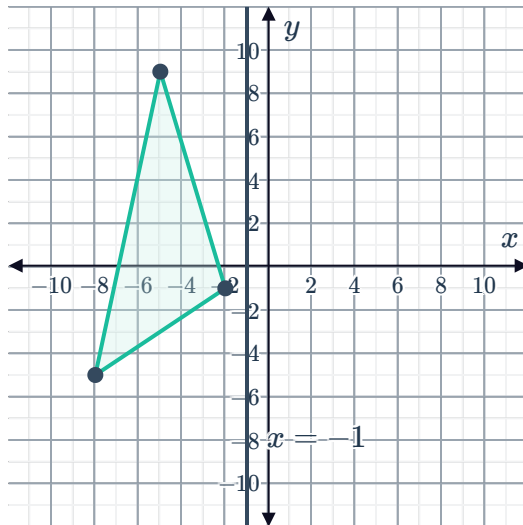


- 1 Plot point A on a coordinate plane.
Then, plot point A' , a reflection of point A across the x -axis.
 - a $A(7, 3)$
 - b $A(-5, -6)$
 - c $A(2, -2)$
 - d $A(-5, 6)$
- 2 Plot point A on a coordinate plane.
Then, plot point A' , a reflection of point A across the y -axis.
 - a $A(6, 9)$.
 - b $A(-7, -3)$
 - c $A(9, -10)$
 - d $A(-7, 8)$
- 3 Consider the following:
 - a Plot the point $A(6, 7)$.
 - b If point A is reflected about the x -axis to produce point B , determine the coordinates of point B .
 - c How many units would point A need to be translated to overlap point B ?
- 4 Consider the point $A(6, 3)$.
 - a Plot point A on the number plane.
 - b If point A is reflected across the y -axis to produce point B , determine the coordinates of point B .
 - c What distance would point A need to be translated to overlap point B ?
- 5 Plot the line segment AB where the endpoints are $A(-4, -2)$ and $B(6, 7)$. Then plot the reflection of the line segment AB across the x -axis.
- 6 Plot the line segment AB , where the endpoints are $A(-6, -1)$ and $B(10, 8)$. Then, plot the reflection of the line segment about the y -axis.

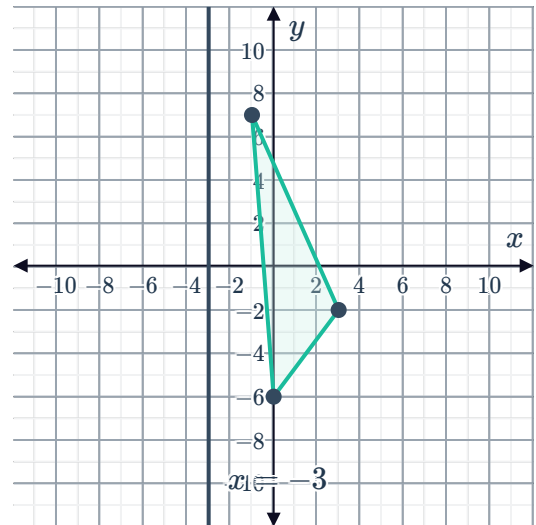
7 Plot the reflection of each of the following triangles across the given line



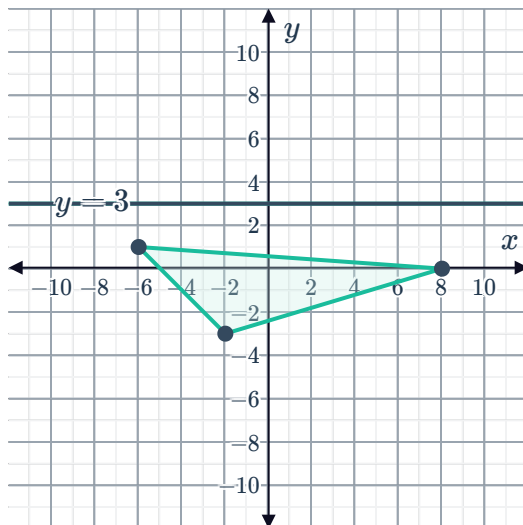
a



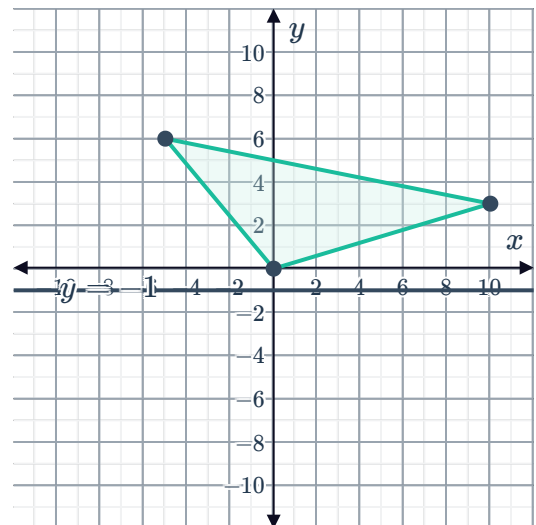
b



c



d

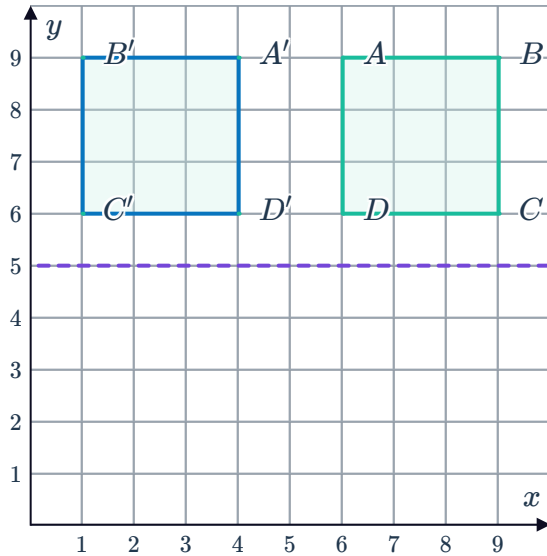


Additional questions

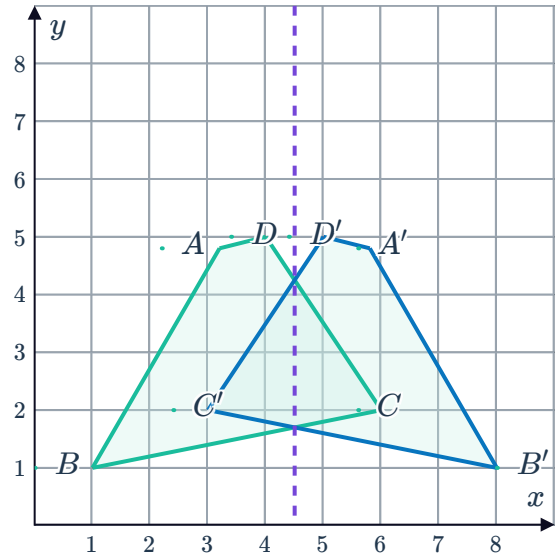
8 Describe the result of reflecting a figure across the x -axis.

- 9 Determine whether each of the following transformations is a reflection across the given line:

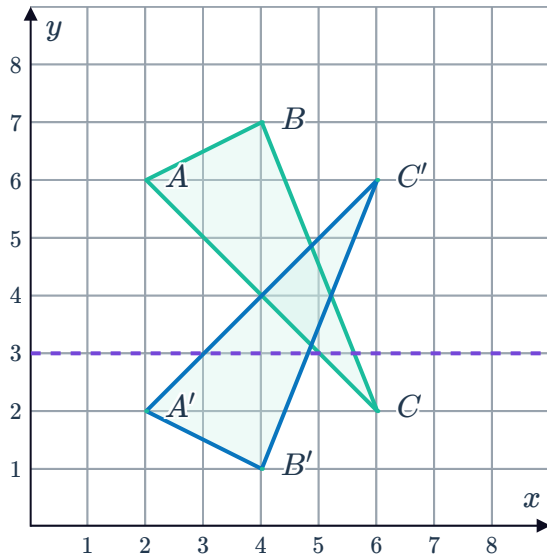
a



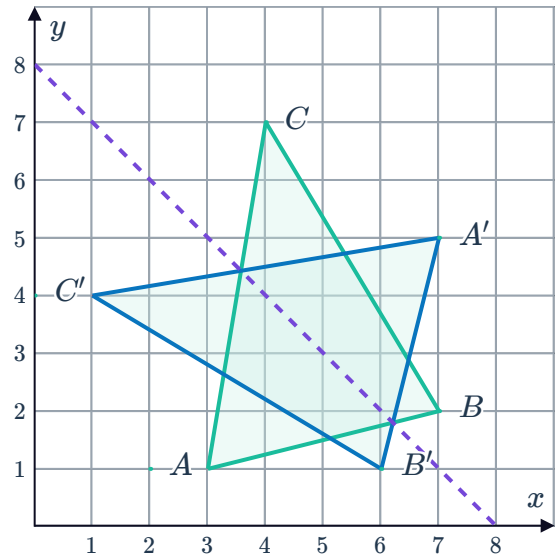
b



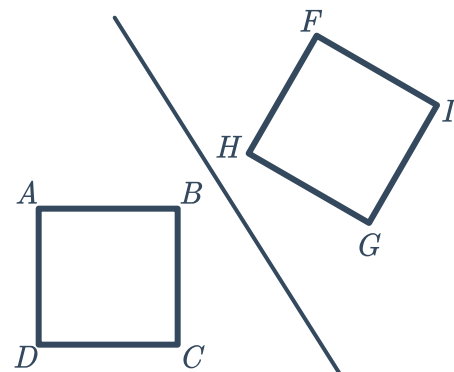
c



d

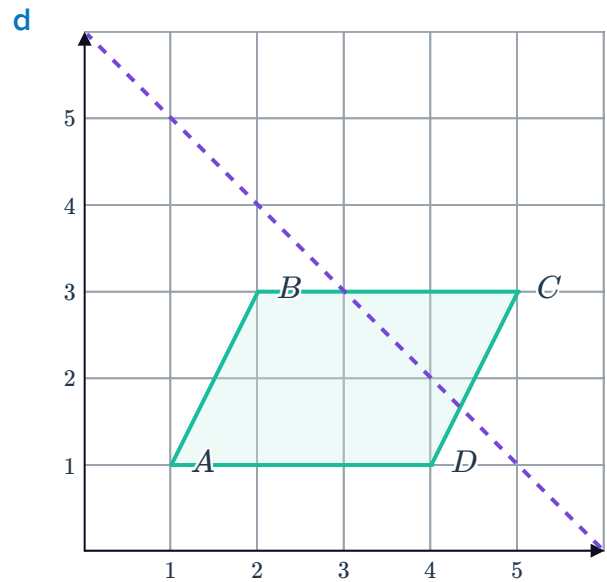
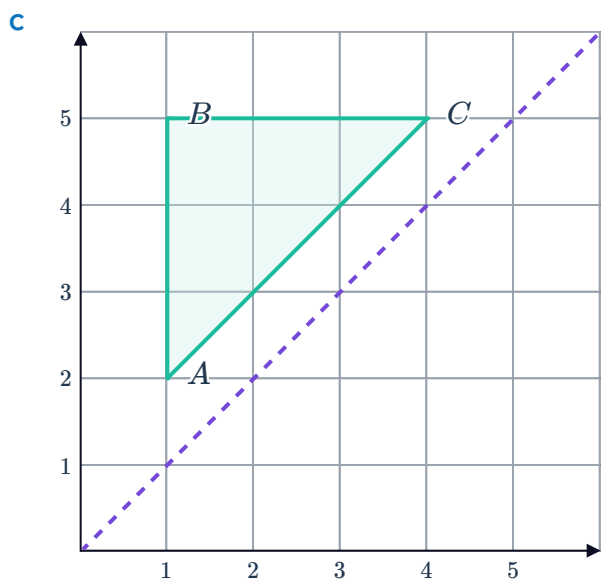
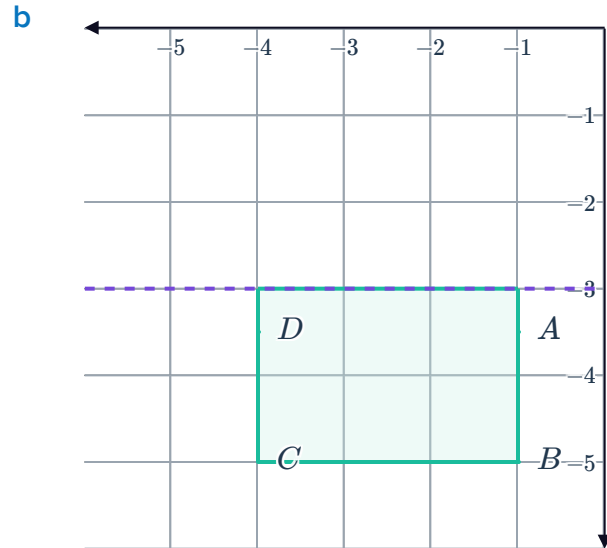
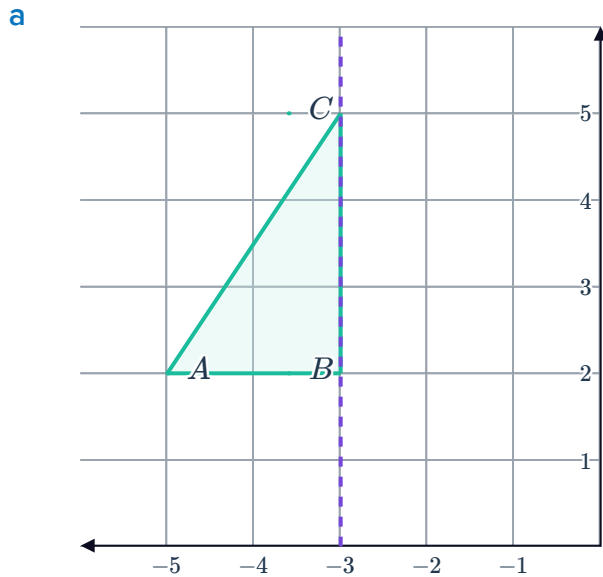


- 10 Square $ABCD$ was reflected across the given line. Which point in the image corresponds to point A ?





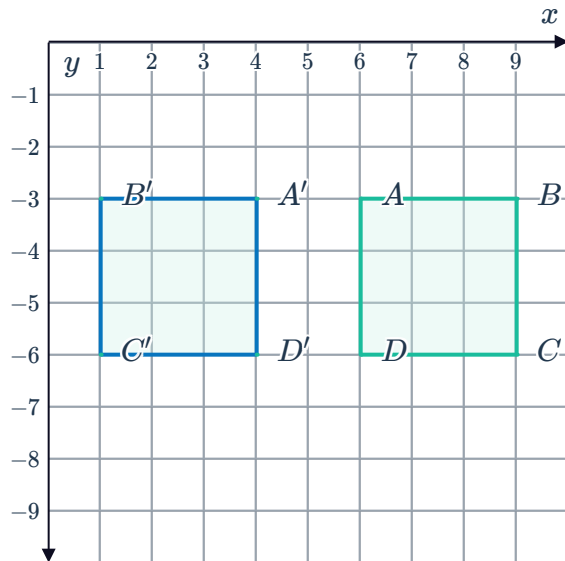
- 11 Describe the characteristics of a figure that has reflectional symmetry and sketch an example.
- 12 Reflect each of the following shapes across the given line of reflection. Label all vertices on the image.



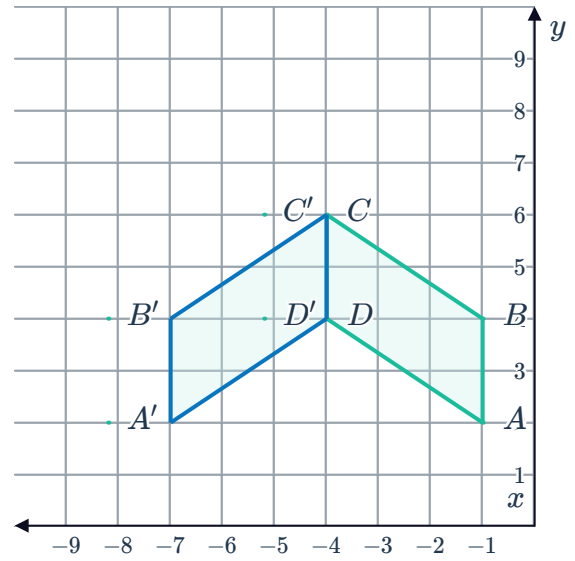
13 Draw the line of reflection for each of the following transformations:



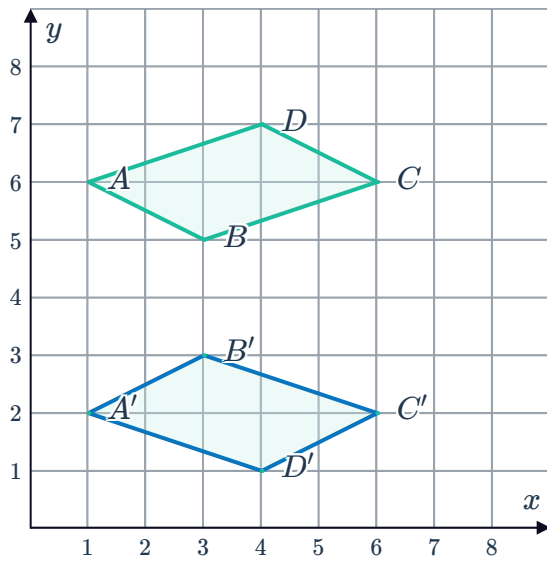
a



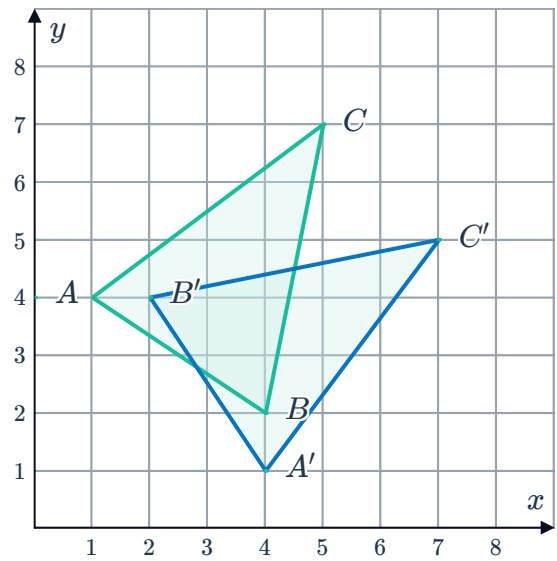
b



c



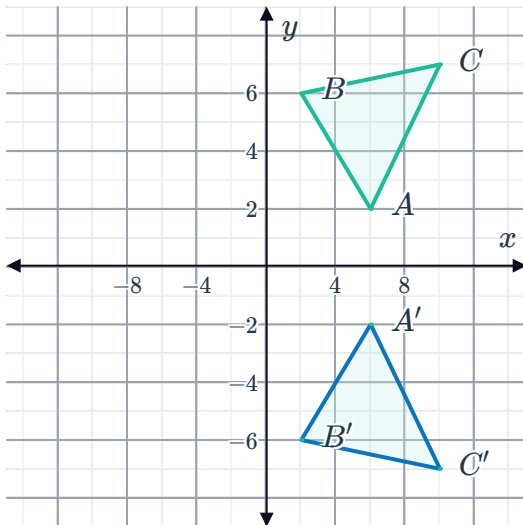
d



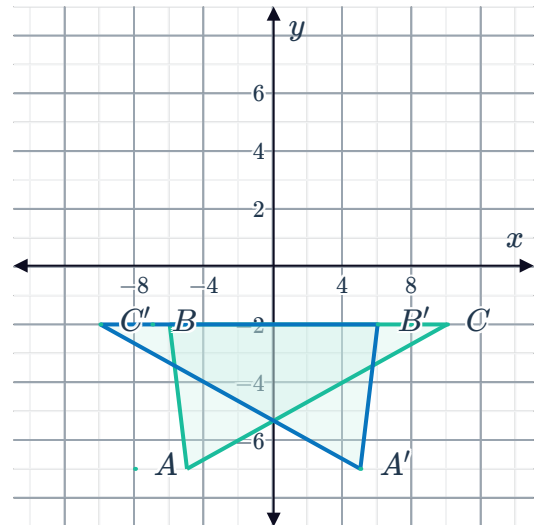
14 Describe each of the following reflections:



a

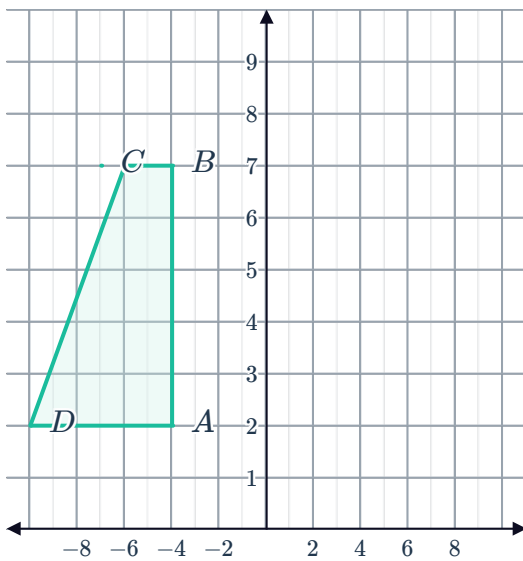


b

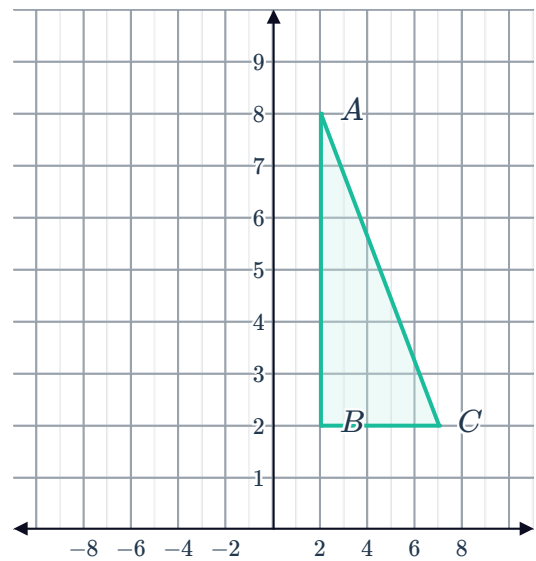


15 Reflect the following shapes across the given the lines of reflection:

a Line of reflection: y -axis



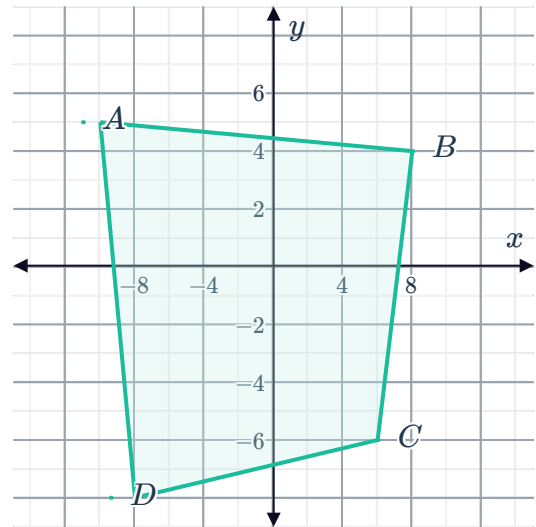
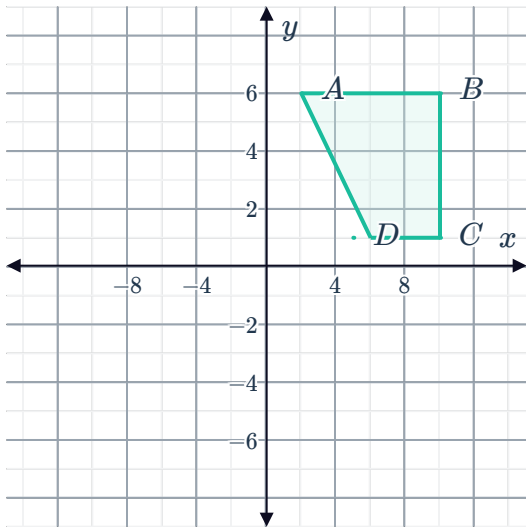
b Line of reflection: y -axis



c Line of reflection: x -axis

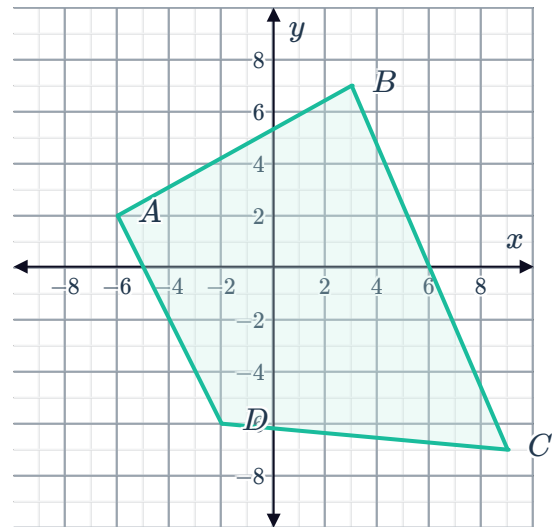
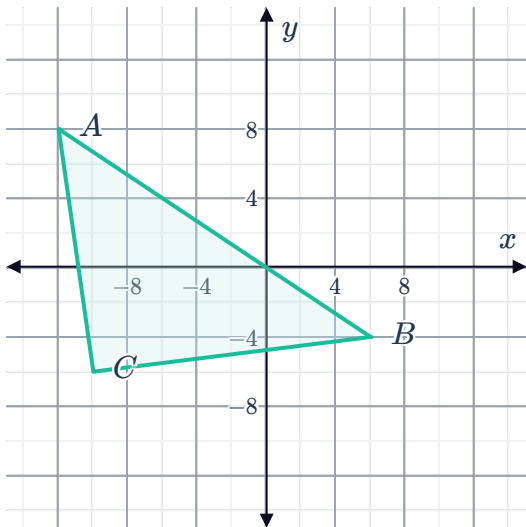


Line of reflection: x -axis



e Line of reflection: $y = x$

f Line of reflection: $y = -x$



16

- a If point $M(5, 6)$ is reflected across the y -axis to produce point N , what are the coordinates of point N ?
- b What distance would the original point M need to be translated to map to point N ?

- 17 If point $F(-3, -7)$ is reflected across the line $y = -x$ to produce point B , what are the coordinates of point B ?

18 Determine whether each of the following statements are true or false:



- a When a figure is reflected across the x -axis, only the sign of the y -value changes.
- b When a figure is reflected across a vertical line, only the y -value changes.
- c When a figure is reflected, the preimage and image have the same size and shape.
- d When a figure is reflected, the preimage and image have the same orientation.
- e A figure is reflected across a line. If the reflected figure is then reflected across the same line again, it will return to the original figure.

19 Sokayna drew figure A and Roly drew figure B .

- a Can Sokayna's triangle be reflected to produce Roly's? If so, describe the reflection.
- b Roly thinks his triangle has a greater perimeter than Sokayna's. Is this true? Explain why or why not.

